could we determine whether they have feelings or whether they are just a collection of mindless algorithms?

When it comes to humans, we are today capable of differentiating between conscious mental experiences and non-conscious brain activities. Though we are far from understanding consciousness, scientists have succeeded in identifying some of its electrochemical signatures. To do so the scientists started with the assumption that whenever humans report that they are conscious of something, they can be believed. Based on this assumption the scientists could then isolate specific brain patterns that appear every time humans report being conscious, but that never appear during unconscious states.

This has allowed the scientists to determine, for example, whether a seemingly vegetative stroke victim has completely lost consciousness, or has merely lost control of his body and speech. If the patient's brain displays the telltale signatures of consciousness, he is probably conscious, even though he cannot move or speak. Indeed, doctors have recently managed to communicate with such patients using fMRI imaging. They ask the patients yes/no questions, telling them to imagine themselves playing tennis if the answer is yes, and to visualise the location of their home if the answer is no. The doctors can then observe how the motor cortex lights up when patients imagine playing tennis (meaning 'yes'), whereas 'no' is indicated by the activation of brain areas responsible for spatial memory.[7]

This is all very well for humans, but what about computers? Since silicon-based computers have very different structures to carbon-based human neural networks, the human signatures of consciousness may not be relevant to them. We seem to be trapped in a vicious circle. Starting with the assumption that we can believe humans when they report that they are conscious, we can identify the signatures of human consciousness, and then use these signatures to 'prove' that humans are indeed conscious. But if an artificial intelligence self-reports that it is conscious, should we just believe it?

So far, we have no good answer to this problem. Already thousands of years ago philosophers realised that there is no way to prove conclusively that anyone other than oneself has a mind.